



Physics-Informed and Explainable Data-Driven Modelling of Minimum Film Thickness in Plain Journal Bearings

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Abstract: This work developed a machine learning (ML) model for predicting the minimum film thickness (h_0) in plain journal bearings. A reference dataset comprising 376,349 datapoints was generated from the approximated pressure journal design method developed by Reason and Narang. The inputs in the dataset included geometric and operational features: Society of Automotive Engineers (SAE) oil grades, speed (500–10,000 rpm), bearing load (1,500–9,500 N), bearing length/diameter (0.01–0.08 m), clearance (0.00001–0.00006 m), and initial oil temperature (40–70 °C). Four supervised learning models, i.e., Artificial Neural Network (h_0 -ANN), Gaussian Process Regression (h_0 -GPR), Ordinary Least Squares Regression (h_0 -OLS), and Ridge Regression (h_0 -Ridge) were developed and evaluated. The results indicated that Artificial Neural Network (ANN) provided the most accurate predictions via achieving an R^2 value of 0.99. The GPR model obtained an R^2 value of 0.98 while those of the OLS and Ridge Regression achieved R^2 of 0.95 and 0.93, respectively. Having shown superior predictive capabilities, the ANN model was selected for further statistical evaluation. Analysis from Local Interpretable Model-agnostic Explanations (LIME), Shapley Additive exPlanations (SHAP), and radar plots all proposed that the ANN model offered stable predictions within the considered parametric space. The model was presented in an explicit mathematical form in an Excel spreadsheet; its computational efficiency was recommended for the design of plain journal bearings.

Keywords: Film thickness; Machine learning; Neural network; Predictions; Bearing

1 Introduction

The minimum film thickness h_0 of a plain journal bearing is the smallest portion of fluid to ensure separation of the journal from the bearing. The film helps the bearing avoid friction and sustains the applied load. Some conditions such as low speeds, higher loads, increased lubricant temperatures, etc. have been noted for the decrease of film thickness [1]. This can lead to increased tear and wear and sometimes critical bearing failure. It is therefore essential for the lubricant film to be maintained at a certain thickness so that the applied load can always be fully supported by the oil film [2].

The applied load is a critical factor that determines how much film thickness is required, especially if operating at high performance when there is a real chance of quickly thinning the fluid film [3]. If the film becomes too thin, local contact pressures can increase significantly, probably causing the lubricant to degrade or fail. There has even been reported cases where the shaft and bearing are welded together as a direct result of the metal contact [4]. This makes it even more essential to accurately calculate the h_0 during the design of bearings.

It is noteworthy that the overall efficiency of the bearing system could help maintain a healthy separation between the two metal surfaces. Friction arising from microscopic surface contours could lead to increased friction, heat generation, and significant energy losses, thus ultimately reducing efficiency [5]. Continuously operating machinery like turbines and compressors has been known to experience a drop in efficiency attributed to issues of lubrication over a period of time [6].

The viscosity of the lubricant is essential when determining the appropriate film thickness and inferring the thermal characteristics of a bearing across several operating conditions [7, 8]. The average operating and initial oil temperature can negatively affect bearing performance when not properly evaluated. It is anticipated that rising temperatures may reduce oil viscosity and potentially lead to a thermal breakdown as well as a compromise of bearing functions.

In modern engineering applications, to produce bearings that offer robustness and are environmentally-friendly, it is vital to study and analyse the h_0 [9]. This warrants a better grasp of wear and tear, develops new lubrication fluids and techniques, and improves system efficiency.

There have been several research on the h_0 of journal bearings and how several operating and geometric parameters affect it. For instance, Soni and Vakharia [10] investigated the effects of friction, torque, applied load, and power loss in short journal bearings operating in thin film and mixed-lubrication regimes. Using an adjusted Reynolds equation which had a viscosity correction, they investigated the influence of molecular layers in very thin oil films. They noted that more realistic predictions were obtained with the effects of the molecular films than with just traditional hydrodynamic models. At high-load and low speed conditions, the scholars observed an improvement in the h_0 . Their work concluded that as the interaction between fluid and metal become more significant, conventional lubrication theory might not effectively explain the system behaviour under mixed lubrication.

Beamish et al. [11], with ultrasonic sensing methods, examined the h_0 under normal, heavy, constantly changing loads, and varying speeds. The study demonstrated that the ultrasonic approach to obtaining non-invasive thickness readings offered satisfactory, reliable, and repeatable results in agreement with experimental data that are notably impractical with conventional measurement methods. The effects of misaligned shafts and a lack of lubrication on the film thickness were captured, matching computational results closely. It was thus clearly shown that the ultrasonic measurement was a practical tool for monitoring bearing lubrication in real time.

Mota et al. [12] simplified the Navier-Stokes equations by reducing them to elliptical differential equations that are pressure-based. Their solutions were numerically obtained using a centred-finite difference technique; the results were adopted to improve modelling lubricant characteristics between concentric cylindrical surfaces. They evaluated equilibrium positions, hydrodynamic forces, pressure profiles, stiffness, and damping responses in various bearing configurations, namely elliptical, cylindrical, and worn. The model showed good matches with existing models and was available on ROSS-Rotordynamics Open Source Python library.

On the other hand, Hong [13] focused on improving the lubrication of misaligned journal bearings impacted by large loads. The researcher examined the effects of steady and unsteady loads on the pressure distribution, h_0 , motion of shaft, and surface deformation of the bearings. The study found that the h_0 had an increase of about five times when a low modulus rubber with flexible grooves were introduced to the bearing, compared with the use of only flexible structures.

In recent years, machine learning (ML) techniques have been employed to infer key engineering systems, thereby making designs smarter, more adaptive, and easily predicting failure. There have been several successes in this regard as in predictive maintenance and prognostics [14–18], structural health monitoring, and damage detection [19–23], thermal and energy system modelling [24–28], modelling mechanical systems [29, 30], computational fluid dynamics (CFD) and flow prediction [31–34], and aerospace systems [35–38].

Keeping up with these trends, some authors have utilised Artificial Intelligence to infer several bearing operational characteristics, including minimum film thickness of journal bearings. The list of such studies has been summarised in Table 1. An overview of the reviewed papers shows that most of the works used Artificial Neural Network (ANN) to model their predictions and base their reference on either experiments or numerical results from the solution of Reynolds equation.

From these studies, ML models in this area have mostly been presented in a manner that facilitates understanding of the hidden-layer components, thus turning the models into a ‘black box’. This gap in ML modelling motivated the current study.

The novelty of this study, therefore, lied in its presentation of the minimum film thickness model in an explainable and interpretable form using local and global statistical tools such as Local Interpretable Model-agnostic Explanations (LIME) and Shapley Additive exPlanations (SHAP), respectively. The model is also developed as a mathematical equation embedded in an Excel spreadsheet that could easily be deployed in the design of bearings. The remaining sections of this work are as follows: Section 2 explains the design of full film hydrodynamic bearings and the ML techniques deployed to predict the minimum film thickness. Section 3 displays the obtained results and discusses them. The paper concludes in Section 4, highlighting key findings and stating the limitations of the model.

Table 1. Summary of recent studies using machine learning (ML) to predict bearing parameters

Author(s)	ML Technique	No. of Datapoints	Source of Dataset	Predicted Bearing Parameter	Evaluation Metrics	Model Explicitly Expressed
Lindley et al. [39]	Gaussian process	Not stated	Experiment	Oil film thickness	Not stated	No
Badawi et al. [40]	ANN and fuzzy logic	Not stated	Simulation from the solution of Reynolds equation	Load-carrying capacity, attitude angle, friction variable, and maximum-film-pressure ratio	Error: <2.5%	No
Saleh et al. [41]	PINNs	35	Electrohydrodynamic simulation results	Pressure and minimum film thickness	Error: Training: Pressure model: 1%, Minimum film thickness: 2%; Testing: Pressure model: 5%, Minimum film thickness: 12%	No
Navthar et al. [42]	ANN	Not stated	Experiment	Minimum film thickness	N/A	No
Baş and Karabacak [43]	ANN, SVM, and RT	168	Experiment	Friction torque and friction coefficient	ANN MSE and R^2 : Friction torque: 0.0012794 and 0.99; Friction coefficient: 0.0000103 and 0.98; SVM MSE and R^2 : Friction torque: 0.075658 and 0.88; Friction coefficient: 0.000803 and 0.84; RT MSE and R^2 : Friction torque: 0.008941 and 0.98; Friction coefficient: 0.000148 and 0.97	No
Cartwright et al. [44]	Neural network	39,000,000	OpenFOAM simulation results from the solution of Reynolds equation	Oil film thickness, temperature, surface deformation, and pressure across the bearing surface	Mean Error (%): Oil film thickness: 0.086, temperature: 0.0036, surface deformation: 0.20 and pressure across the bearing surface: 0.39	No
Ji et al. [45]	ANN, KNN and RF	Not stated	Experiment	Temperature	ANN: R^2 : 0.97, RMSE: 1.906, MAE: 1.196; KNN: R^2 : 0.98, RMSE: 1.386, MAE: 0.618; RF: R^2 : 0.97, RMSE: 1.934, MAE: 1.254	No
Sinanoglu et al. [46]	ANN	Not stated	Experiment	Pressure variation	Not stated	No
Ghorbanian et al. [47]	ANN	1,858	Finite-difference solution of Reynolds equation	Friction loss, oil consumption, minimum oil film thickness, eccentricity, oil temperature rise and displacement relative to shell	R^2 : Friction loss: 0.99, oil consumption: 0.99, minimum oil film thickness: 0.99, eccentricity: 0.99, displacement relative to shell: 1.00	No
Kumar et al. [48]	ANN	Not stated	Data from literature	Minimum film thickness	Error: <0.3%	No
Zhao and Wong [49]	PINN	Not stated	Data from the solution of Reynolds equation	Film thickness	Relative error <6%	No

Note: ANN = Artificial Neural Network; SVM = Support Vector Machine; RT = Regression Trees; KNN = k nearest neighbours; RF = Random Forest; PINNs = Physics-Informed Neural Networks; MSE = mean square error; RMSE = root mean square error; MAE = mean absolute error.

2 Methodology

2.1 Approximate Method for the Design of Full Film Hydrodynamic Bearing

The distribution of pressure generated by the motion of a plain journal bearing as shown in Figure 1 was determined by solving the Reynolds equation, which is a reduced form of the Navier-Stokes fluid flow equations. The steady form of the Reynolds equation was given in Eq. (1):

$$\frac{\downarrow h^3}{\bar{x} \rightarrow \mu} \frac{p}{x} + \frac{\downarrow h^3}{\bar{z} \rightarrow \mu} \frac{p}{z} = 6V \frac{h}{x} + 6h \frac{V}{x} \quad (1)$$

where, x , y , and z are rectangular coordinates (m); h is film thickness (m); μ is dynamic viscosity (Pa-s); p is fluid pressure (Pa) and V is journal velocity (m/s).

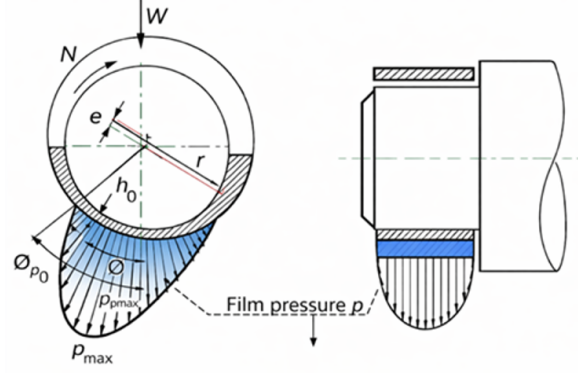


Figure 1. Motion generating pressure in fluid between the journal and the bearing

The Reynolds equation may be evaluated through either approximate analytical formulations or numerical computation. Following the determination of the hydrodynamic pressure field, the operational performance of the journal bearing could be quantified in terms of load support, frictional power loss, requirements of lubricant supply, and the associated temperature rise within the lubricant film. As an alternative to directly solving Eq. (1), designers frequently rely on a set of empirical design charts developed by Raimondi and Boyd [50–52], who applied an iterative solution methodology to the Reynolds equations. These charts correlate the Sommerfeld number with critical bearing characteristics, including distribution of film thickness, friction coefficient, lubricant flow rate, side-flow ratio, the location of minimum film thickness, the magnitude and position of maximum pressure, and the film rupture or termination angle.

To eliminate time spent on using design charts, the bearing design method developed by Reason and Narang [53] was employed in this study. The pressure variation approximation of this method was given by Eq. (2):

$$p = \frac{\frac{3\mu V}{\pi^2} \frac{L^2}{4} - z^2 \frac{\varepsilon \sin \theta}{(1+\varepsilon \cos \theta)^3}}{1 + \frac{2+\varepsilon^2}{2r^2} \frac{L^2}{4} - z^2 \frac{1}{(1+\varepsilon \cos \theta)(2+\varepsilon \cos \theta)}} \quad (2)$$

where, μ is viscosity (Pa·s); V is surface velocity (m/s); r is journal radius (m); c is radial clearance (m); L is bearing length (m); z is axial coordinate (m); ε is eccentricity variable; and θ is circumferential angular coordinate of the bearing circumferential direction.

The solution obtained from the integration of Eq. (2) is presented in terms of sine (I_s) and cosine (I_c) integrals. Both integrals are functions of the eccentricity ratio (ε) and L/D ratio. The bearing parameter-minimum film thickness was simulated in using the MATLAB R2023a software, resulting in 376,349 datapoints. The procedure is summarised in the flowchart as shown in Figure 2. The procedure involves the use of a viscosity chart and interpolations for sine and cosine integrals and the Sommerfeld number. Therefore, this study experimented other methods, specifically ML techniques, to eliminate the use of charts and thus reducing computational time.

2.2 Gaussian Process Regression

Gaussian Process Regression (GPR) is a non-parametric and probabilistic ML technique grounded in Bayesian inference and widely employed for nonlinear regression problems in engineering and physics. In GPR, the target variable is modelled as a realization of a Gaussian process, fully characterized by a mean function and a covariance (kernel) function. The kernel encodes prior assumptions about the smoothness, continuity, and correlation structure of the underlying physical process, to enable GPR to learn complex input–output relationships directly from data while providing predictive uncertainty estimates.

In the present work, model training and prediction simulations were performed in Python using the scikit-learn ML library [54]. GPR was employed to predict the minimum film thickness h_0 using seven input variables like Society of Automotive Engineers (SAE) oil grades, speed, bearing load, bearing length/diameter, clearance, and initial oil temperature. These inputs represent the governing operating and geometric conditions influencing h_0 . Prior to model training, the input features were standardized to maintain numerical stability and balanced kernel length scales. The dataset was split into training, validation, and testing sets in a 70:15:15 ratio to ascertain that the model was properly evaluated and to reduce the risk of overfitting.

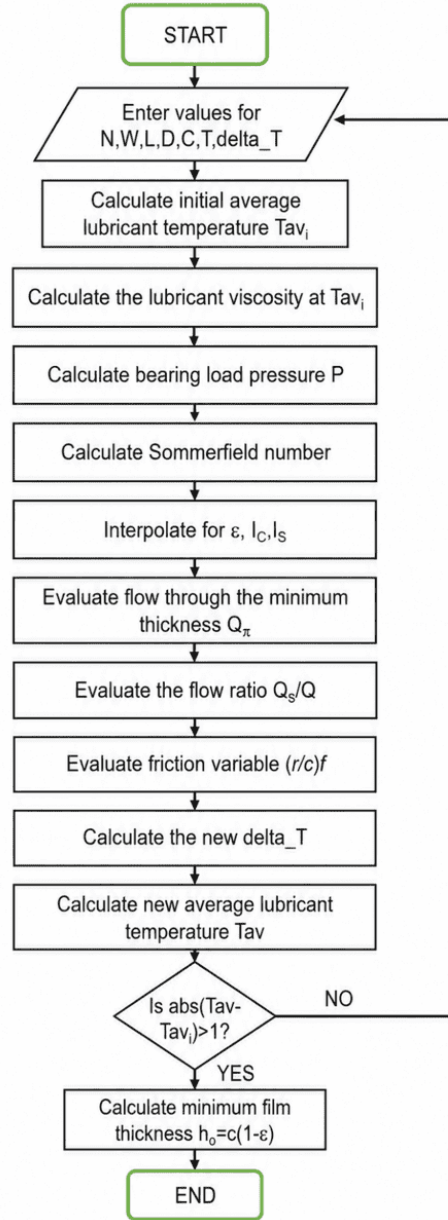


Figure 2. Flowchart for evaluating the minimum film thickness

Traditional GPR could become too slow for large datasets because inverting the kernel matrix was computationally expensive. To overcome this, a scalable approximation using Random Fourier Features (RFF) was employed [55]. This method approximates the radial basis function (RBF) kernel in a finite feature space, turning the GPR problem into a simpler Bayesian linear regression task. The flexibility of GPR is maintained even though the training is on a large dataset. Model performance is then assessed using R^2 , mean square error (MSE), and root mean square error (RMSE) to quantify prediction accuracy.

2.3 Ordinary Least Squares

Ordinary Least Squares (OLS) Regression is a conventional method used to model the relationship between a dependent variable and one or more independent variables. In OLS, the target is expressed as a linear combination of the input features plus an error term, and the model finds the parameters that minimize the squared difference between observed and predicted values.

Because it is simple, interpretable, and grounded in solid theory, OLS was often used as a baseline model in data-driven studies and engineering applications [56, 57]. Mathematically, the OLS model can be written as:

$$y = X\beta + \varepsilon \quad (3)$$

where, y is the vector of observed output, X is the design matrix containing the input variables, β is the vector of regression coefficients, and ε represents the random error term. The optimal coefficient vector is obtained by solving the normal equations,

$$\beta = (X^T X)^{-1} X^T y \quad (4)$$

which yields the best linear unbiased estimator under the assumptions of linearity, independence, homoscedasticity, and normally distributed errors [58]. In the present study, OLS Regression was used to predict the output parameter h_0 from the physics-informed synthetic input variables utilising the scikit-learn library. Prior to model fitting, the input features were standardized to guarantee numerical stability and to prevent dominance of variables with larger magnitudes. The dataset was randomly divided into training, validation, and testing subsets to enable robust performance evaluation. Model accuracy was quantified using R^2 , MSE, and RMSE.

2.4 Ridge Regression

The Ridge Regression algorithm with cross-validation (RidgeCV) was employed in this study to predict the minimum film thickness h_0 . The RidgeCV model was implemented using the scikit-learn library in Python. Ridge Regression is a regularized linear regression technique that mitigates multicollinearity and overfitting by introducing an L_2 -norm penalty on the regression coefficients.

In Ridge Regression, the model parameters are obtained by minimizing the following objective function:

$$\min_{\beta} \left(\sum_{i=1}^n (y_i - x_i^T \beta)^2 + \alpha \|\beta\|_2^2 \right) \quad (5)$$

where, y_i denotes the measured value of h_0 , x_i represents the input feature vector, β is the vector of regression coefficients, and α is the regularization parameter controlling the trade-off between data fidelity and smoothness of the model.

To capture nonlinear dependencies between the input variables and h_0 , polynomial feature expansion of degree two was applied prior to model training. All input features were standardized to zero mean and unit variance to safeguard numerical stability and balanced regularization. The optimal value of the regularization parameter α was automatically determined using k -fold cross-validation (with $k = 5$) on the training dataset. The selection criterion was based on maximizing the coefficient of determination, R^2 . This procedure secured robust generalization while preventing overfitting.

3 Results and Discussion

3.1 Synthetic Data

This was generated using the method developed by Reason and Narang [53], following the steps outlined in the flowchart of Figure 2. There were a total of 376,349 datapoints comprising 7 inputs, including oil grade (SAE), speed in rpm, load in N, bearing length in m, bearing diameter in m, bearing clearance in m, and initial oil temperature in °C, while the minimum film thickness in mm was the evaluated parameter.

3.1.1 Data normalisation

Data normalization is a crucial preprocessing procedure in ML, employed to scale input and output variables to a comparable range. This process prevents features with larger numerical magnitudes from disproportionately influencing the training process, thereby enhancing both model accuracy and generalization. In this study, the dataset presented in Table 2 revealed that the maximum value of the speed variable was approximately eight orders of magnitude greater than that of the minimum film thickness. Such a disparity implied that, without normalization, speed could dominate the learning process and bias the prediction of h_0 . To ensure all input features contribute fairly to the model, min-max normalization was applied as shown in Eq. (6). This technique rescaled each feature to a standard range, usually between -1 and 1 , while preserving the relative differences between datapoints intact. Without this step, features with larger values could dominate the learning process, rendering it harder for the model to learn properly. Normalizing the data also helped the model train faster and make the optimization stabler.

$$X_n = 2 \left(\frac{X - X_{\min}}{X_{\max} - X_{\min}} \right) - 1 \quad (6)$$

where, X represents the original value of the variable, $X_{\min}$ and $X_{\max}$ are the minimum and maximum values of that variable in the dataset, and X_n represents the normalized value after scaling. The normalisation of each input parameter is presented as equation A1 to A8 in the supplementary materials of Appendix A.

Table 2. Statistical overview of synthetic data

	Oil Grade	Speed (rpm)	Load (N)	Length (m)	Diameter (m)	Clearance (m)	Initial Oil Temperature (°C)	h_0 (m)
Min.	10	500	1,500	0.01	0.01	0.00001	40	1.01E-07
Max.	60	10,000	9,500	0.08	0.08	0.00006	70	5.39E-05
Mean	32.26476	2,914.736	5,201.442	0.041906	0.038011	0.0000414	54.86668	1.1E-05
Std.	16.55589	2,906.857	2,573.726	0.020876	0.019342	0.0000149	10.98682	1.11E-05

3.2 Machine Learning Model Performance

The predictive performance of the four ML models encompassed Artificial Neural Network (h_0 -ANN), Gaussian Process Regression (h_0 -GPR), Ordinary Least Squares Regression (h_0 -OLS), and Ridge Regression (h_0 -Ridge). Data were randomly partitioned using MATLAB's divider and function with a 70:15:15 training-validation-test ratio. The random seed was not explicitly fixed, resulting in training (263,445 samples), validation (56,452 samples), and testing (56,452 samples) as shown in Table 3. Across all metrics and datasets, the h_0 -ANN obviously performed the best.

Table 3. Evaluation of the machine learning (ML) techniques under study

	h_0 -ANN			h_0 -GPR			h_0 -OLS			h_0 -Ridge		
	R^2	MSE	RMSE	R^2	MSE	RMSE	R^2	MSE	RMSE	R^2	MSE	RMSE
Training	0.99	0.00022	0.0148	0.98	0.005	0.07	0.95	0.006	0.077	0.93	0.009	0.094
Validation	0.99	0.00021	0.0145	0.98	0.004	0.06	0.94	0.007	0.083	0.94	0.008	0.089
Testing	0.99	0.00021	0.0145	0.98	0.004	0.06	0.95	0.006	0.077	0.93	0.008	0.089

Note: ANN = Artificial Neural Network; GPR = Gaussian Process Regression; OLS = Ordinary Least Squares Regression; MSE = mean square error; RMSE = root mean square error.

The neural network with architecture [7-30-1] achieved R^2 values of about 0.99 for training, validation, and testing, meaning it captured almost all the variations in the output. Its MSE values were small (around $2.1\text{--}2.2 \times 10^{-4}$), resulting in RMSE values between 0.0145 and 0.0148. The fact that these numbers were so consistent across all datasets suggested that the model learned the patterns appropriately and did not overfit.

The GPR model had a close performance to that of ANN, with R^2 around 0.98, but its errors were higher: MSE ranged from 0.004 to 0.005 and RMSE from 0.06 to 0.07. This showed that while GPR could handle nonlinear relationships, it was not as accurate as the neural network model. The slight drop in performance could be due to kernel choice or the challenges of scaling GPR to such a large dataset.

The two linear models, h_0 -OLS and h_0 -Ridge, lagged in predictive abilities. Their test R^2 values were between 0.95 and 0.93, respectively, and their errors were noticeably larger. Both linear models struggled to capture the complex nonlinear patterns that the neural network handled easily. These results highlighted the apparent advantage of the neural network in modelling the minimum film thickness. It did not only fit the data well but also effectively generalized to new cases.

3.2.1 Justifications for selecting minimum film thickness_Artificial Neural Network (h_0 -ANN)

The ANN model was trained with a maximum limit of 1,000 epochs and terminated after 487 epochs when the validation-based early stopping criterion was reached. Six consecutive checks without further improvement in validation performance signalled the early stop. During training, the MSE decreased from an initial value of 10.9 to 2.2×10^{-4} , while the gradient was reduced from 17.9 to 1.95×10^{-4} , demonstrating effective convergence of the network. Based on the consistently high R^2 values, the lowest MSE and RMSE, and the strong agreement between training, validation, and testing results, h_0 -ANN was selected as the optimal predictive model. Its superior ability to learn complex nonlinear mappings, combined with robust generalization performance, turned it into the most reliable and accurate technique among those evaluated for modelling the target variable. The trends shown in Figure 3 further substantiate the quantitative evaluation provided by the R^2 , MSE, and RMSE metrics. Across all operating conditions, the h_0 -ANN predictions closely followed the reference solution of Reason and Narang [53], to accurately capture the expected physical behaviour: increasing minimum film thickness with higher oil viscosity grade, and decreasing film thickness with increasing temperature, clearance, and reduced speed. h_0 -GPR generally captured the overall trends but started to deviate at the extreme of the operating conditions, which were in accordance with its higher RMSE values. The linear models, h_0 -OLS and h_0 -Ridge, consistently over or under-predicted when

the data became strongly nonlinear. That explained why their R^2 scores were lower and their errors were higher. h_0 -ANN not only kept the overall error to a minimum but also followed the correct physical trends across the full range of conditions. The inference abilities of the h_0 -ANN model was further assessed using its cross plot as shown in Figure 4. The predicted values exhibited a tight clustering around the 45° line of equality, indicating excellent agreement with the actual targets across the full range of the dataset. The absence of systematic bias and the minimal scatter about the diagonal further demonstrated consistency and robustness of the model. The final network configuration comprised a single hidden layer with 30 neurons. These observations are fully aligned with the high R^2 and low RMSE values reported earlier, hence reinforcing the reliability and strength of generalizing the model.

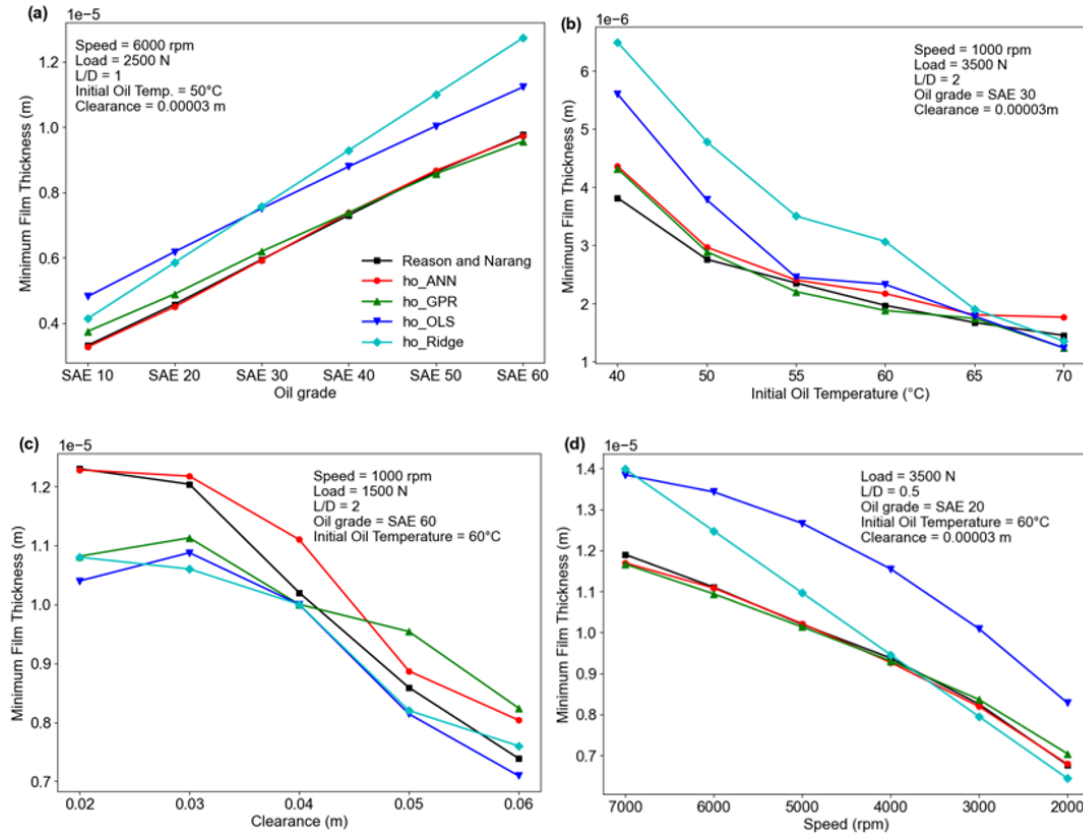


Figure 3. Comparison of the machine learning (ML) techniques

Note: h_0 -ANN = Artificial Neural Network model for predicting minimum film thickness; h_0 -GPR = Gaussian Process Regression model for predicting minimum film thickness; h_0 -OLS = Ordinary Least Squares Regression model for predicting minimum film thickness; h_0 -Ridge = Ridge Regression model for predicting minimum film thickness.

3.3 Explainability and Interpretability of the Model

LIME is an interpretability technique designed to explain individual predictions of complex ML models. It operates by perturbing input features in the local vicinity of a prediction as well as fitting a simple and interpretable surrogate model to approximate the behaviour of the complex model locally. By providing feature-level contributions for specific instances, LIME enhances transparency, supports local reasoning, and improves trust in black-box models without requiring access to their internal structure.

The plot shown in Figure 5 provides a local and instance-specific explanation of the model prediction for the minimum oil film thickness under the operating conditions of SAE 40 oil, 1,000 rpm speed, 4,500 N load, and specified bearing geometry. The most dominant negative contribution arises from bearing length ≤ 0.02 m. For short journal bearings, axial side leakage significantly reduces pressure buildup, hence limiting the hydrodynamic load-carrying capacity and consequently decreasing the minimum film thickness. This behaviour is well-documented in finite-length bearing theory and Reynolds equation solutions as evidenced in the works of Pinkus and Sternlicht [59]. The rotational speed shows a negative local contribution, which aligns with the work of Dowson [60]. While increasing speed usually helps thicken the lubricant film, complex interactions with load and clearance tend to reduce this benefit at higher speeds, especially when viscous heating or geometric limitations come into play. The LIME plot highlights this non-linear behaviour. There is a positive contribution from the diameter, and this confirms the physical trend where by larger diameters boost surface velocity and increase hydrodynamic pressure, leading to

thicker minimum films [50–52]. On the other hand, load shows a negative contribution, following the well-known inverse relationship between applied load and film thickness with higher loads reducing separation between surfaces. At the prescribed conditions, oil grade and initial oil temperature also contribute positively. This matches classical lubrication theory, in which higher viscosity and lower temperature help maintain a thicker oil film. Basically, viscosity controls how effectively the lubricant could generate hydrodynamic pressure in the converging wedge between the shaft and bearing.

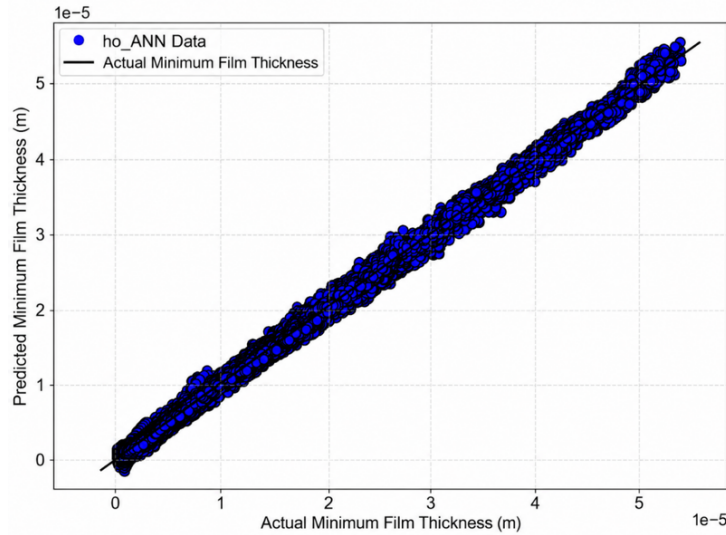


Figure 4. Cross plot of the selected model, Artificial Neural Network (ANN)

Note: h_0 -ANN = Artificial Neural Network model for predicting minimum film thickness.

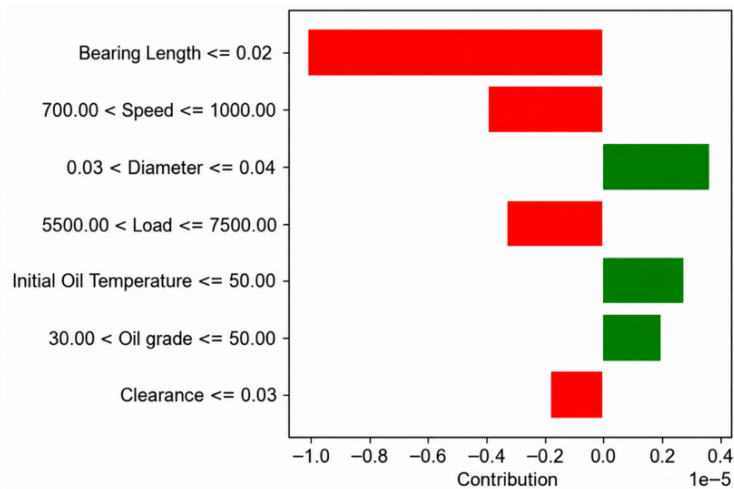


Figure 5. Local Interpretable Model-agnostic Explanations (LIME) feature contribution plot for the selected operating conditions

Note: Green bars denote positive contributions that increase the model prediction whereas red bars denote negative contributions that decrease it.

Finally, the negative contribution associated with small radial clearance indicates a localized reduction in the effectiveness of hydrodynamic wedge formation at the considered operating point. Although clearance is a critical parameter for establishing pressure generation in journal bearings, excessively small clearances could become detrimental, particularly in short or finite-length bearings as in the case considered here. In such cases, the lubricant film has limited space to develop a converging wedge, and axial side leakage further suppresses pressure buildup. As a result, the minimum film thickness may decrease despite the reduced clearance.

Therefore, the negative LIME contribution of clearance in this instance was interpreted as a local effect, reflecting the specific combination of short bearing length, moderate-to-high load, and operating speed. This behaviour reinforces the importance of selecting an optimal clearance rather than a minimum clearance, and it validates that the model is capturing physically meaningful and nonlinear interactions among geometric parameters.

The radar (spider) plots in Figure 6 juxtapose the h_0 -ANN predictions with the reference model from Reason and Narang [53] across different operating conditions. In Figure 6a, the model was tested across lubricant grades (SAE 10–SAE 60), and in Figure 6b, it is shown under different loads (2,000–7,000 N). In both cases, the ANN predictions closely followed the reference model. The dashed red lines almost perfectly overlap the solid blue ones; therefore, the model is capturing the nonlinear behaviour of the system accurately.

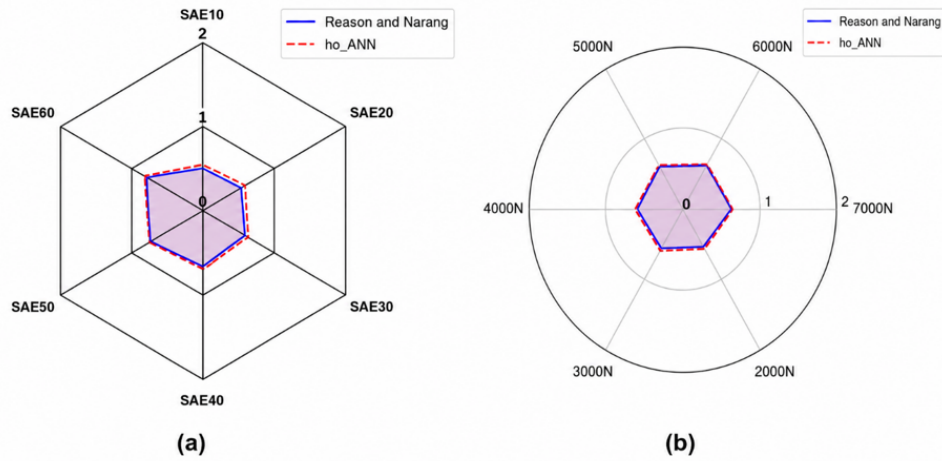


Figure 6. Comparison of reference and predicted values at (a) various oil grades: 4,000 rpm, 3,500 N, aspect ratio (L/D) = 2, clearance = 0.00003 m, 60 °C and (b) varying loads: Society of Automotive Engineers (SAE) 30, L/D = 1, clearance = 0.00003 m, 40 °C

Note: The degree of overlap between the two profiles indicates the level of agreement between the models, whereas deviations between the curves highlight differences in the predicted and actual values.

Across all the conditions in the spider plots, there is a significant consistency in the ANN predictions. It is evident that the area and overall shape of the polygons remain stable; this is a sign that the model is robust and generalizable. The few deviations therein are systematic, reflecting the fact that there are not many random errors and no overfitting.

Finally, these plots clearly explain the choice of the h_0 -ANN model. It does not just match the reference results but also provides a reliable and data-driven alternative that could handle complex interactions on its own, without detailed physical assumptions. In other words, it is accurate, consistent, and flexible, to comply with the exact expectations of a predictive model.

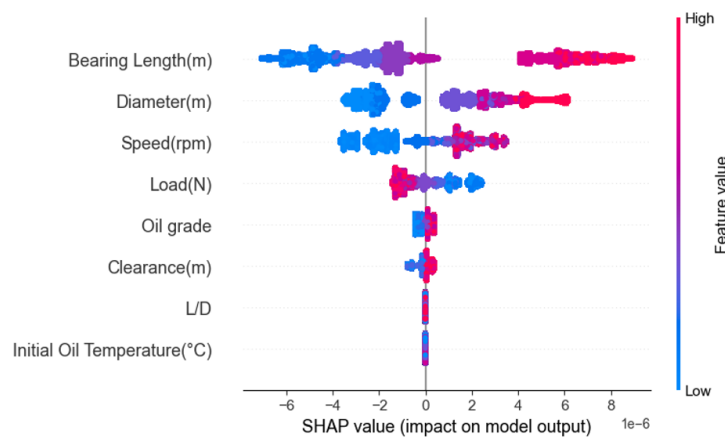


Figure 7. Impact of each input parameter and their Shapley Additive exPlanations (SHAP) values

Note: Red markers correspond to high feature values and blue markers correspond to low feature values, while the SHAP value on the x-axis quantifies the direction and magnitude of each feature's impact on the prediction.

A SHAP plot is widely used to graphically show and enable the understanding of the impact of each input parameter on the target variable. The average positive impacts of each feature were emphasized after having assessed all possible combinations of parameters. It thus provides a global view of which features matter most and whether

their influence increases or decreases the predicted output, compared with a baseline value. In Figure 7, the colour scheme helps clarify these trends: red points represent higher feature values, while blue points represent lower ones. The horizontal spread of the points illustrates how strongly each input feature affects the minimum film thickness, e.g., wider spreads indicate a larger impact. Features with larger impact mean absolute SHAP values (positioned toward the top) exerted a stronger effect on the prediction. Within the parametric space of the prediction, bearing length, diameter, speed, and load exhibited the most pronounced contributions to the prediction, indicating that geometric scale and operating conditions dominantly governed minimum film thickness. These findings are consistent with classical electrohydrodynamic and hydrodynamic lubrication literature, reporting that film thickness increases with bearing size and operating speed while decreasing with higher loads. For example, bearing geometry and load relationships have been shown to significantly affect film thickness in journal lubrication studies. Viscosity and lubricant properties are likewise established contributors to film formation in electrohydrodynamic contexts. In addition, effects of bearing length on film thickness have been documented in engine bearing analyses [61].

3.4 Tolerance of the Artificial Neural Network Algorithm to Measurement Uncertainty

The robustness of the h_0 -ANN algorithm to input errors was further quantified by repeatedly introducing some random errors into the seven input predictors. An input error of $X\%$ was deployed to the dataset by multiplying each input data by a random number between 1 and $1 \pm X\%$. To obtain the variations represented as error bars, each case was repeated severally to account for the random nature of the perturbations. This continued until the mean and rms coefficient of determination converged. The mean of the R^2 and the rms variation of the R^2 is presented in Eq. (7) and Eq. (8).

$$\hat{\mu}_N = \frac{1}{N} \sum_{i=1}^N R_i^2 \quad (7)$$

$$\hat{\sigma}_N = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (R_i^2 - \hat{\mu}_N)^2} \quad (8)$$

where, N is the total number of repeated runs, μ_N is the mean value of the R^2 results after N runs, i refers to the index of the run, going from 1 to N , R_i^2 is the coefficient of determination obtained from the i -th run and σ_N is the standard deviation of the R^2 values after N runs. Figure 8 shows that the predictions of the model are robust to input errors because the mean R^2 stayed above 0.9 for errors as high as 20%.

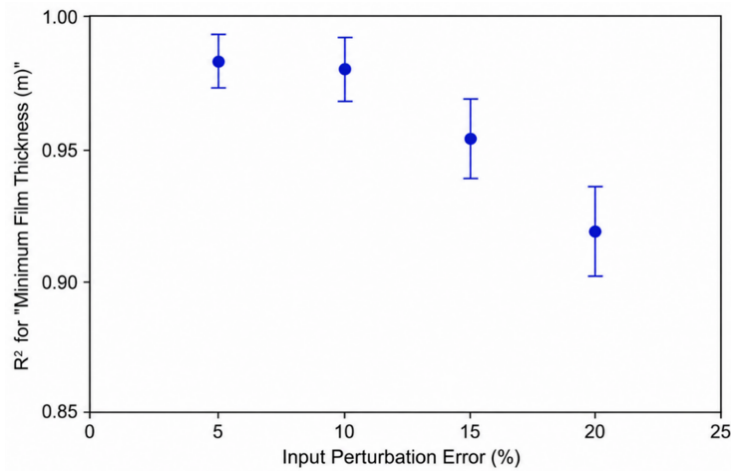


Figure 8. Coefficient of determination of the Artificial Neural Network (ANN) model as input errors to the dataset

3.5 Mathematical Description of the Minimum Film Thickness Artificial Neural Network (h_0 -ANN) Model

This model established using the Levenberg-Marquardt technique was expressed in Eq. (9).

$$h_0 = \sum_{j=1}^{30} \left\{ \text{purelin} \left[LW_{j,1} \left(\sum_{i=1}^{29} \sum_{j=1}^{30} \text{tansig} (X_i * IW_{i,j} + b_1) \right) \right] + b_2 \right\} \quad (9)$$

where, the functions `purelin` and `tansig` correspond to fundamental transfer functions in MATLAB employed to generate network predictions from the input variables. The `tansig` function represents the hyperbolic tangent sigmoid activation function, which introduces nonlinearity into the hidden layer. In contrast, `purelin` denotes a linear activation function that establishes a linear mapping between the hidden-layer output and the final network output.

The variable X_i represents the model input parameters, SAE oil grades, speed, bearing load, bearing length/diameter, clearance, and initial oil temperature. The matrix IW contains the weights connecting the input layer to the hidden layer, while LW denotes the weights linking the hidden layer to the output layer. The terms b_1 and b_2 represent bias vectors associated with the hidden and output layers, respectively, which were used to adjust the neuron activation levels and improve model fitting.

The normalised equation for the model was expressed in Eq. (10).

$$h_{0_{\text{norm}}} = \sum_{i=1}^{30} X_i - 0.18 \quad (10)$$

Finally, after de-normalisation, the final model could be expressed as in Eq. (11).

$$h_0 = \frac{(5.37 \times 10^{-5} \times h_{0_{\text{norm}}}) + (2 \times 1 \times 10^{-7}) + 5.37 \times 10^{-5}}{2} \quad (11)$$

A full breakdown of all the weights and biases in the model build-up is presented in Appendix A. A ready-to-use Excel spreadsheet with the model formula has been attached in the supplementary material.

3.6 Inference Latency of the Model

Inference latency is simply the amount of time a trained neural network takes to produce a prediction after receiving an input. This is a critical factor in applications that require fast or real-time responses, such as embedded systems, control applications, or live monitoring tools.

Inference latency depends on several practical factors including the structure of the network, the activation functions used, the numerical precision of computations, and the efficiency of the software and hardware. This modelling work was executed on a 13th Gen Intel® Core™ i7-1355U processor with a frequency of 1.7 GHz which corresponds to 1.7×10^9 clock cycles per second. The time taken per clock cycle was estimated to be 588 picoseconds per clock cycle. An approximate value of 5-clock cycles per neuron was allocated to the tanh function used to smoothen nonlinearity [62]. Latency was estimated analytically from the processor clock frequency rather than a direct measurement in the hardware.

The architecture of the ANN model is [7-30-1] which translates to an aggregation function of $[(7 \times 30) + (30 \times 1) + (30 + 1)]$ giving 271 doubles or 2,168 bytes. The weights perform 240 operations; biases are 31 operations and the output layer needs 5 cycles, creating a total of 155 cycles for the activation function.

For a single prediction to be made using the h_0 -ANN model, it will require:

$$240(\text{weights}) + 31(\text{biases}) + 155(\text{activations}) = 426 \text{ cycles}$$

Therefore, the time required will be:

$$426 \times 588 \text{ ps} \approx 250.5 \text{ ns}$$

This extremely low latency demonstrated that the proposed neural network model was appropriate for real-time deployment, including applications such as online combustion monitoring, control, or surrogate modelling within CFD. The compact architecture and low computational complexity permit rapid inference while maintaining sufficient predictive capability.

4 Conclusions

1. Both h_0 -OLS and h_0 -Ridge are linear regression models and they show lower predictive strength, which corresponds to an explained variance as low as 93%. This shows that the underlying process contains nonlinear interactions not fully captured by linear models. Although h_0 -GPR produced strong results, with R^2 values 0.98 to confirm its suitability for nonlinear modelling, comparing it with h_0 -ANN, GPR showed higher error values (RMSE), suggesting that while it is effective, h_0 -ANN model produces more accurate predictions under the studied conditions.

2. The h_0 -ANN model demonstrates robustness and high generalizability strength, having had its inputs perturbed with varying errors up to 20% and obtaining mean R^2 values that stayed above 0.9.

3. The h_0 -ANN model has been explicitly presented as a mathematical equation in Excel format and could make inferences in 250.5 nanoseconds, which reduces the experimental procedure timescale by over 50%.

4. LIME and SHAP plots have been deployed to locally and globally assess the accuracy of the developed model, proving its suitability for deployment in bearing design.

5. Bearing length and diameter are the dominant predictors of minimum film thickness of a plain journal bearing. This implies the aspect ratio (L/D) is a vital design consideration during the design of bearings.

Limitations of the Study and Future Work

The proposed ANN model was developed and evaluated using synthetic dataset generated from the reference bearing model, thus enabling a detailed coverage of the operating conditions considered in this study. It is worth noting that although the results showed excellent predictive accuracy within the parametric space considered in this study, the performance of the model when applied to industrial measurements might be influenced by factors not fully captured in the synthetic data, such as material property variations, sensor uncertainties, tolerances, and operating-condition fluctuations. Nevertheless, the use of physics-based synthetic data provides a robust foundation for model development and preliminary assessment. Future work will focus on validating the developed model using experimental and industrial datasets to assess its generalization capability under real-world conditions.

Author Contributions

Conceptualization, N.A.; methodology, N.A. and O.E.A.; software, N.A. and I.U.M.; validation, N.A., O.A., and O.E.A.; formal analysis, N.A. and O.A.; investigation, N.A. and I.U.M.; resources, N.A. and I.U.M.; data curation, O.E.A.; writing—original draft preparation, N.A.; writing—review and editing, N.A. and I.U.M.; visualization, O.A.; supervision, O.E.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Nomenclature

ANN	Artificial Neural Network
CFD	computational fluid dynamics
GPR	Gaussian Process Regression
h_0	minimum film thickness
IW	input weight
LIME	local interpretable model-agnostic explanations
LM	Levenberg–Marquardt
LW	line weight
MSE	mean square error
OLS	Ordinary Least Squares
R^2	coefficient of determination
RBF	radial basis function
RFF	Random Fourier Features
RMSE	root mean square error
RT	Regression Trees
SHAP	Shapley Additive exPlanations
SVM	Support Vector Machine

Appendix A

Normalisation of Variables

$$\text{Oil grade}_n = 2 \left(\frac{\text{Oilgrade} - 10}{50} \right) - 1 \quad (\text{A1})$$

$$\text{Speed}_n = 2 \left(\frac{\text{Speed}-500}{9500} \right) - 1 \quad (\text{A2})$$

$$\text{Load}_n = 2 \left(\frac{\text{Load} - 1500}{8000} \right) - 1 \quad (\text{A3})$$

$$\text{Length}_n = 2 \left(\frac{\text{Length} - 0.01}{0.07} \right) - 1 \quad (\text{A4})$$

$$\text{Diameter}_n = 2 \left(\frac{\text{Diameter} - 0.01}{0.07} \right) - 1 \quad (\text{A5})$$

$$\text{Clearance}_n = 2 \left(\frac{\text{Clearance} - 0.00001}{0.00005} \right) - 1 \quad (\text{A6})$$

$$\text{Initial Oil Temperature}_n = 2 \left(\frac{\text{Initial Oil Temperature} - 40}{70} \right) - 1 \quad (\text{A7})$$

$$\text{Minimum Film Thickness}_n = 2 \left(\frac{\text{Minimum Film Thickness} - 0.0000001009}{0.0000538} \right) - 1 \quad (\text{A8})$$

Weights and Biases of the Neurons in the Minimum Film Thickness Artificial Neural Network (h_0 -ANN) Model

$$X_1 = 0.2066 * \tanh((0.329696 * \text{Oilgrade}) + (2.57268 * \text{Speed}) - (0.18289 * \text{Load}) - (0.32554 * \text{Length}) + (0.40285 * \text{Diameter}) - (0.67989 * \text{Clearance}) - (0.27325 * \text{InitialOilTemp})) + 3.1909$$

$$X_2 = 0.066 * \tanh((0.77083 * \text{Oilgrade}) - (0.93548 * \text{Speed}) + (2.264082 * \text{Load}) - (6.3814 * \text{Length}) + (7.046535 * \text{Diameter}) + (0.242938 * \text{Clearance}) + (0.557479 * \text{InitialOilTemp})) + 5.160$$

$$X_3 = 0.3413 * \tanh((0.216727 * \text{Oilgrade}) + (0.472686 * \text{Speed}) - (0.38816 * \text{Load}) + (0.929144 * \text{Length}) + (0.453104 * \text{Diameter}) - (0.90758 * \text{Clearance}) - (0.11291 * \text{InitialOilTemp})) - 0.1411$$

$$X_4 = -10.02 * \tanh((-0.13729 * \text{Oilgrade}) - (0.05588 * \text{Speed}) + (0.11729 * \text{Load}) - (0.24873 * \text{Length}) - (0.22729 * \text{Diameter}) + (0.1689 * \text{Clearance}) + (0.062074 * \text{InitialOilTemp})) - 0.489$$

$$X_5 = -5.602 * \tanh((0.391664 * \text{Oilgrade}) + (1.414179 * \text{Speed}) - (0.42106 * \text{Load}) + (0.169826 * \text{Length}) + (1.48281 * \text{Diameter}) - (0.53149 * \text{Clearance}) - (0.2779 * \text{InitialOilTemp})) + 2.079$$

$$X_6 = -8.118 * \tanh((-0.23027 * \text{Oilgrade}) - (0.35069 * \text{Speed}) + (0.2623 * \text{Load}) - (0.85511 * \text{Length}) - (0.29328 * \text{Diameter}) + (0.268555 * \text{Clearance}) + (0.159934 * \text{InitialOilTemp})) + 0.418$$

$$X_7 = -1.7132 * \tanh((-0.3046 * \text{Oilgrade}) - (0.571 * \text{Speed}) + (0.731868 * \text{Load}) - (1.09231 * \text{Length}) - (0.38859 * \text{Diameter}) + (0.34899 * \text{Clearance}) + (0.20557 * \text{InitialOilTemp})) + 0.420$$

$$X_8 = 11.57 * \tanh((0.26603 * \text{Oilgrade}) + (0.184901 * \text{Speed}) - (0.70624 * \text{Load}) + (1.17628 * \text{Length}) + (1.240124 * \text{Diameter}) - (0.34801 * \text{Clearance}) - (0.2004 * \text{InitialOilTemp})) + 0.663$$

$$X_9 = -5.432 * \tanh((-0.15987 * \text{Oilgrade}) - (0.23563 * \text{Speed}) + (0.122185 * \text{Load}) - (0.11875 * \text{Length}) - (0.21956 * \text{Diameter}) - (0.06098 * \text{Clearance}) + (0.147438 * \text{InitialOilTemp})) + 0.0372$$

$$X_{10} = -7.525 * \tanh((0.34409 * \text{Oilgrade}) - (4.0611 * \text{Speed}) - (0.56345 * \text{Load}) + (1.016042 * \text{Length}) + (0.371862 * \text{Diameter}) - (0.00296 * \text{Clearance}) - (0.18921 * \text{InitialOilTemp})) - 5.93$$

$$X_{11} = 4.209 * \tanh((0.094091 * \text{Oilgrade}) - (0.12951 * \text{Speed}) + (0.10098 * \text{Load}) + (0.729508 * \text{Length}) + (0.353645 * \text{Diameter}) - (0.24422 * \text{Clearance}) - (0.05881 * \text{InitialOilTemp})) - 0.237$$

$$X_{12} = 5.708 * \tanh((0.085652 * \text{Oilgrade}) + (0.192138 * \text{Speed}) - (0.09364 * \text{Load}) + (0.363943 * \text{Length}) + (0.010798 * \text{Diameter}) + (0.125338 * \text{Clearance}) - (0.09152 * \text{InitialOilTemp})) - 0.178$$

$$X_{13} = -0.332 * \tanh((0.376454 * \text{Oilgrade}) + (0.344698 * \text{Speed}) - (0.46305 * \text{Load}) - (0.54427 * \text{Length}) + (1.973144 * \text{Diameter}) - (0.49938 * \text{Clearance}) - (0.26934 * \text{InitialOilTemp})) - 0.095$$

$$X_{14} = -0.138 * \tanh((0.551917 * \text{Oilgrade}) + (0.594605 * \text{Speed}) - (0.29579 * \text{Load}) + (3.878225 * \text{Length}) - (4.08799 * \text{Diameter}) + (0.068342 * \text{Clearance}) - (0.35669 * \text{InitialOilTemp})) - 2.800$$

$$X_{15} = 0.195 * \tanh((0.531793 * \text{Oilgrade}) + (1.299285 * \text{Speed}) - (0.29969 * \text{Load}) + (0.072476 * \text{Length}) + (0.348805 * \text{Diameter}) - (0.52201 * \text{Clearance}) - (0.27618 * \text{InitialOilTemp})) + 1.322$$

$$X_{16} = -7.246 * \tanh((0.095845 * \text{Oilgrade}) - (0.08503 * \text{Speed}) + (0.023975 * \text{Load}) + (0.677807 * \text{Length}) + (0.327326 * \text{Diameter}) - (0.22822 * \text{Clearance}) - (0.06123 * \text{InitialOilTemp})) - 0.229$$

$$X_{17} = 1.600 * \tanh((-0.6007 * \text{Oilgrade}) - (0.80799 * \text{Speed}) + (2.116865 * \text{Load}) - (0.85843 * \text{Length}) - (0.86734 * \text{Diameter}) + (0.489868 * \text{Clearance}) + (0.344024 * \text{InitialOilTemp})) + 1.855$$

$$X_{18} = 11.86 * \tanh((-0.12741 * \text{Oilgrade}) - (0.18038 * \text{Speed}) + (0.106393 * \text{Load}) - (0.28062 * \text{Length}) - (0.12356 * \text{Diameter}) - (0.0897 * \text{Clearance}) + (0.124198 * \text{InitialOilTemp})) + 0.04$$

$$X_{19} = 5.388 * \tanh((-0.24859 * \text{Oilgrade}) - (0.13228 * \text{Speed}) + (0.67967 * \text{Load}) - (1.16683 * \text{Length}) - (1.33138 * \text{Diameter}) + (0.360296 * \text{Clearance}) + (0.191794 * \text{InitialOilTemp})) - 0.601$$

$$X_{20} = -6.750 * \tanh((0.140555 * \text{Oilgrade}) + (0.017053 * \text{Speed}) - (0.02718 * \text{Load}) - (0.05958 * \text{Length}) - (0.01697 * \text{Diameter}) - (0.11427 * \text{Clearance}) - (0.03578 * \text{InitialOilTemp})) - 0.0411$$

$$X_{21} = -5.268 * \tanh((-0.38702 * \text{Oilgrade}) - (1.45576 * \text{Speed}) + (0.416221 * \text{Load}) - (0.14661 * \text{Length}) - (1.50897 * \text{Diameter}) + (0.539674 * \text{Clearance}) + (0.275719 * \text{InitialOilTemp})) - 2.0793$$

$$X_{22} = 8.591 * \tanh((-0.08323 * \text{Oilgrade}) - (0.12493 * \text{Speed}) + (0.082166 * \text{Load}) - (0.15149 * \text{Length}) - (0.16533 * \text{Diameter}) + (0.106481 * \text{Clearance}) - (0.00297 * \text{InitialOilTemp})) - 0.448$$

$$\begin{aligned}
X_{23} &= 6.231 * \tanh((-0.29081 * \text{Oilgrade}) - (0.20313 * \text{Speed}) + (0.758227 * \text{Load}) - (1.20788 * \text{Length}) - \\
&\quad (1.17505 * \text{Diameter}) + (0.339644 * \text{Clearance}) + (0.212935 * \text{InitialOilTemp})) - 0.725 \\
X_{24} &= -8.511 * \tanh((0.219446 * \text{Oilgrade}) + (0.452099 * \text{Speed}) - (0.37047 * \text{Load}) + (0.868004 * \text{Length}) + \\
&\quad (0.300601 * \text{Diameter}) - (0.30979 * \text{Clearance}) - (0.15426 * \text{InitialOilTemp})) - 0.443 \\
X_{25} &= -7.584 * \tanh((-0.38145 * \text{Oilgrade}) + (3.844124 * \text{Speed}) + (0.606671 * \text{Load}) - (1.08747 * \text{Length}) \\
&\quad - (0.40996 * \text{Diameter}) + (0.060992 * \text{Clearance}) + (0.210857 * \text{InitialOilTemp})) + 5.832 \\
X_{26} &= 2.791 * \tanh((0.22251 * \text{Oilgrade}) + (0.085358 * \text{Speed}) - (0.02227 * \text{Load}) + (0.006291 * \text{Length}) + \\
&\quad (0.055775 * \text{Diameter}) - (0.03307 * \text{Clearance}) + (0.044178 * \text{InitialOilTemp})) + 0.139 \\
X_{27} &= 1.623 * \tanh((0.572519 * \text{Oilgrade}) + (0.782593 * \text{Speed}) - (2.15944 * \text{Load}) + (0.92364 * \text{Length}) + \\
&\quad (0.829421 * \text{Diameter}) - (0.42274 * \text{Clearance}) - (0.33681 * \text{InitialOilTemp})) - 1.879 \\
X_{28} &= -0.4574 * \tanh((-0.37152 * \text{Oilgrade}) + (0.182487 * \text{Speed}) + (0.814111 * \text{Load}) - (1.26537 * \text{Length}) \\
&\quad - (1.25232 * \text{Diameter}) + (0.400097 * \text{Clearance}) + (0.253057 * \text{InitialOilTemp})) - 0.653 \\
X_{29} &= 0.280 * \tanh((0.300479 * \text{Oilgrade}) + (0.391462 * \text{Speed}) - (3.43786 * \text{Load}) + (0.762287 * \text{Length}) + \\
&\quad (0.462712 * \text{Diameter}) + (0.313368 * \text{Clearance}) - (0.17269 * \text{InitialOilTemp})) - 4.322 \\
X_{30} &= -0.891 * \tanh((0.13475 * \text{Oilgrade}) - (0.34623 * \text{Speed}) - (0.14405 * \text{Load}) + (0.599284 * \text{Length}) + \\
&\quad (0.476604 * \text{Diameter}) + (0.088051 * \text{Clearance}) - (0.09075 * \text{InitialOilTemp})) + 1.464
\end{aligned}$$